Conditioning — Bike session, once a week

Not a training session (Stage 0 §2.7) · hip, knee, ankle at loading stage 1 · 12 weeks · 15–39 min

Muscle group

Legs · bike Legs · court

Before: lower-body mobility routine, pre half · then A
From your latest test (B): E = last POS notch · H1 = one notch below the first NEG · H2 = first NEG
Talk test: POS = talking is comfortable · EQ = not sure · NEG = not comfortable
Notch = lever position counted from the lightest end (number them with tape) · HR band = its test HR ±5
Beat 90 = a metronome, one right-foot push per beat · warm-up and cool-down outside the exercise time (R19)

A — Warm-up (10 min · every week · outside the exercise time)

1 **Easy spin with the weekly check stage (cycling)** — 10 min
3 min easy → 4 min at notch E, beat 90: log HR (last 60 s) + talk test
→ 3 min easy with 2 × 20 s quicker spins at a light notch
Test weeks (1, 5, 9, 13): 5 min easy only, then B
notch E · beat 90
Bike · metronome

B — Test (week 1 = your first session, then 5, 9 and 13 · in place of C · ≤28 min)

2 **Talk-test step test (cycling)** — 3-min stages
Beat 90 · same easy start notch every test · +1 notch every 3 min
Last 30 s of a stage: read the same 30-word text aloud → POS · EQ · NEG
Stop at the first NEG (or on R9) → 60 s easy spin, log HR → cool-down
stop at the first NEG
HR monitor · metronome

C — Main set by week (seated · beat 90 · 3 min easy between intervals and rounds)

3 **Base intervals at notch H1 — weeks 2, 3, 4 (cycling)** — 3–4 × 3–4 min
Week 2: 3 × 3 min · week 3: 3 × 4 min · week 4: 4 × 4 min
From minute 2: HR in the H1 band · end: talk "not sure" · 2+ min left
≥2 min in reserve
start at notch H1

4 **Aerobic-power intervals at notch H2 — weeks 6, 7, 8 (cycling)** — 4–5 × 4 min
Week 6: 4 × 4 min · week 7: 5 × 4 min · week 8: 5 × 4 min
From minute 2: HR in the H2 band · end: a few words only · 1+ min left
≥1 min in reserve
start at notch H2

5 **H2 intervals, then badminton rounds — weeks 10, 11, 12 (cycling)** — 4 min + rounds
Wk 10: 3 × 4 min + 2 rounds · wk 11: 3 × 4 + 3 rounds · wk 12: 2 × 4 + 4
Round = 8 × (10 s quick / 20 s easy) on a timer app · HR cap: H2 band + 5
Quick = faster than the beat, smooth, not all-out · a ragged rep ends it
≥1 min in reserve
start at notch H2

D — Court walk-through (weeks 9–12 · ~10 min · before A)

6 **Court walk-through: footwork routes at walking pace** — 10 min
Walk to each corner and back to base, racket up in the ready position
No hop or split-step (R13) · no lunge: both feet share the load (R12)
No swing (racket arm at stage 2) · step around, never pivot on one foot
walking pace only
Court · racket

After: 5 min easy spin or a walk · mobility post half · day after: no badminton
Quads, glute max and calves work hard in the intervals — badminton's day-before muscles (R21)
Archery: no limit — no archery muscle works here; the hands rest light on the bars
Foam rolling: quads and calves from the lower-body menu · next morning: log the step gate (§2.6)

Every session — hypermobility and effort rules

- Saddle so the knee stays bent 25–30° with the pedal at the bottom; never stand on the pedals (R2)
- Sit upright, back neutral (spine at stage 1); hands light, soft elbows, no tight grip (R2)
- Spin, don't grind: can't keep up with the beat → one notch lighter (knee load)
- Skip the session if joints feel loose; stop the moment a joint feels unstable or hurts (R9)
- HR band first, talk test as the check; if they disagree, ride the easier one; never to exhaustion
- One step a week as the cards list; notch settings change only at a test (weeks 1, 5, 9, 13)
- Step gate next morning: a fail repeats that week; two fails in 4 weeks → one step back for 2 weeks (R43)
- Test days: same bike, indoors, same time of day, same kind of day before; usual sleep, food and drink
- Bike taken: walk the easy parts; ride outdoors for training only, never for a test (heat shifts HR)

Session timeline
Mobility pre half → (weeks 9–12: D) → A warm-up → B test or C main set → cool-down → mobility post half
Exercise time (R40: work + rests between) · weeks 2–4: 15 · 18 · 25 min · weeks 6–8: 25 · 32 · 32 min
Weeks 10–12: 32 · 39 · 39 min · test weeks ≤28 min · under ~60 min every week (R19)

Self-evaluation — weeks 5, 9 and 13 against week 1

Gain if · log each test here	Week 13 verdict
EQ or NEG ≥1 notch higher	Gains + clean joints → next period progresses
HR at week 1's EQ notch ≥3% lower	Flat + clean → more intensity or a 2nd session
Check stage (card 1): HR trending down	Any flare → back to week 4's dose
HR recovery at 60 s: log only	Hip, knee, ankle at stage 4 → ask for court drills
Joints: no failed gate, no R9 stop	

v2 — no bike display needed: lever notches, a 90 beat and HR bands; the test counts notches